Seaborn Visualising Continuity with Lines Transcript

Now let us try to see how to plot visual continuity with Siba. Okay, we will be using line plot and replot commands. Yeah, so why do we need to visualise continuity, right? Generally when we look about in a business environment then we are trying to tell a story there is naturally the element of time that comes into play where we want to understand how something is changed over a period of time.

The very moment you bring time it is understood there is a continuous change you want a smooth curve that is representing categories or connecting time periods over a period of time and then the values change. So you can see the trend whether it is increasing, decreasing or increasing at increasing rate, decreasing at decreasing rate so on and so forth. So the right and most intuitive tool for this is connecting data points to revealing the trends and patterns and Seaborn has an inherent advantage because it has transfers the simple line drawing into a statistical pool helping you to build more accurate predictions or building credibility to the data that you are bringing on to the visualisation.

Okay, so now key functions that are used for this purpose are line plot and then you have the replot which helps you to do the subplots within the same category so that you can compare the within the same data for different categories so that you can compare the trends across categories of variables that you have then you can come to a conclusion very easily. We'll see all these examples so that you will be able to understand their use and practise them later. So again I am just we have already imported these libraries I predict yes we have already imported this library so there's no need to import them again.

We are taking an existing data called fMRI data set which is there in the website so we are just downloading that so we are defining fMRI underscore DF as SNS load data set is a default data set that is there in the Seaborn library so we are just taking it and then trying to understand how does it look. Subject, time point, event, region and signal so it is some this is signal crossing data so you are getting the event and you are trying to see the signal okay. So core function what is that you want you want to understand the line plot this is just a simple data we just are using it to showcase certain characteristics of the line plot and replot just to understand you have the subject s1 to s30 then you have the time point which is a continuous variable then you have the signal which is outcome then you have event and region which are categorical variables which can be used for classifying the data.

So now let us try to plot a simple line right and how does this line plot work so line plot has a core function that plots a relationship between X and Y from the data frame and it automatically aggregates Y values that exist for a single X value and gives average for example at a given point in time you have 10 values it would aggregate them and present as an average and the line meaning is it is like for each X value what is the corresponding average value of the X and there is a confidence interval around that line so that you would know the dispersion how much is the mean value and whether it is statistically close bound or it is wide in range so that you can have an understanding of the underlying data also. So simple and just first setting up the plot the figure I'm just setting up the figure so I just got the figure with axis then I am using the SNS line plot command to plot the line so again I am passing the fMRI data frame into data X is equal to time point which is the time interval then Y is equal to signal which is the outcome so I'm not using other objects I'm just using these two and then so you have 0 to 0.5 the time then this is how the signal has varied and it is like this for every time point there might be multiple for example time point 18 in the very short head that we have taken we will take 10 right and I show you that so if you take 10 data points you see so many data points are there at 18 the time point 18 represents a single point on x-axis for which there are so many corresponding values of Y which is signal so the particular point that is shown here represents the average of those 10 values or 7 values for that particular X point and this is automatically done by C1 and the blue translucent blue band around this line indicates the confidence interval with which the mean can fall in this range and you will know that in this particular point the mean has a wider range indicating the underlying data as a wider dispersion than at this point where the translucent band is shorter in width indicating the data is more packed at this particular time point me indicating that the underlying Y values are more closely located than the Y values which are at this particular point so this kind of inferences can be drawn from observing this line plot because there is a confidence interval band around it then we are just adding the titles right so just to improve the aesthetics so again this is the math plot lip functioning over and above the C1 function so you have done that so you got this time point signal strength and average fMRI signal over time right so this is what you have done then let us see a very important characteristic called adding another dimension with human state so this is this is very interesting to understand the use of human state so let us say for example the line plot that we have seen in this particular graph right the line plot which we have seen in this graph is talking only about time point and signal where are there are other three characteristics in the data set like the subject event and region so what if you want to bring those characteristics on to this graph without changing the basic nature of the graph I still want to have a line graph but I want to showcase all this so we will use the hue and style options which are the features which you use for overlaying other dimensions on the existing graph and bringing richness to your visualisation okay so so the main purpose of using human style is to compare trends between subgroups within the same plot and a few parameters separates data into different lines or different in this case lines are different bars are different components based on the colour right and each unique value gets a unique colour using the hue so each category can represent a different colour using the hue parameter and style parameter separates data into different lines because this is a line part based on the line pattern based on solid dashed and dotted if it is a bar graph it would produce different types of bar for each categories so therefore if you use combined style and hue it allows much more rich dimensional presentation and which are very much useful in presenting condensate data on to one particular plot bringing more richness into the graphic so but there is again a catch here if you are trying to print the graphic right as long as you are showing it in a digital environment it should not matter much but if you are presenting it in a let us say in a black and white print out the options of hue and style might be limited still you can work them out with black and white and grey colourings but they would not be as vivid as the colours that you would see on a digital environment so I just keep that in mind while trying to make the use of hue and style so let us take an example so we are just plotting if okay let us take this all to the new code and I will explain this so I'm just plt.figure produces a figure size of 12 by 7 then I am doing sns.lineplot my data is the fMRIDF X is the time point Y is the dull which is nothing but essentially this the same graph so the same code is presented here sns.lineplot but what is that I am trying to do I am trying to add region as hue so for each region I should get a different colour and each event should have a different style and there are markers and dashes to ensure that each data point is represented and the titles remain the same and I am running the code so what is happening so this is the thing so for example you have two types of in peritoneal and frontal event right so you have the peritoneal events which are again the peritoneal region and frontal region right so you have two different types of regions one is represented by blue another is represented by orange and then these are again represented by two different events in terms of the style graph one is a dotted line line with dots in it and another is line with Q the line the event which is representing Q is given by line with a cross whereas the event representing stimulus and Q it is dark line with a dot in between so you are able to see this right let us say what is this particular blue line representing it is representing peritoneal region and it is a stimulus event because you have a thick line with dot now as compared to this you have the dashed line with a cross again it is a peritoneal region which is the Q right so you can see the Q signal is lower as compared to the stimulus signal for both peritoneal and frontal events whereas peritoneal events on an average of higher signal strength compared to the frontal events okay so these conclusions you can draw and compared to this particular graph which is a single line representing the strength at each time point here you are now having the signal strength represented by not only the region but also by the event in the single graph bringing much more richness into this graph okay yeah so creating subplots right we can also do this and achieve this with the subplots okay so we can do this with these subplots so we will use the faceting feature on it is something like the small samples that we have used in the small multiples that we have used in the power be environment where based on a particular categorical feature you would re-plot the same graphic again and again in the same space so that you can visually compare them in the same location right so let us try to see how this faceting will work okay at the right tool is the re-plot and in this case we are using kind is equal to line we can also use bar depending upon the use case okay so structuring is the column variable and row variable that we have to define we will look at this call variable creates a new column for the plot for unique category and creates okay if you use the variable as a column you get the plots as a column if you use the variable in row you get it in rows so that's how your facets are arranged let us look it by taking at the code taking the code to the environment so that becomes easier so what am I doing instead of SNS line plot here I am using the code as re-plot please look at the difference here I am looking at the line plot I am just using the re-plot data is same the data frame of X variable is time point signal you still have few event right the column region is event okay the column this is the main variable of interest for us for every region that you have you want to have the different graph and you want to to be represented them as two different vertical columns or three different vertical columns depending upon number of categories are there in the particular region variable and the kind of the graphic that you want is like and you have given the characteristics for this and then you gave the subtitle fMRI signal comparison across brain regions right and yeah so then you are trying to show the graphic so simply what is happening so you have the peritoneal region the same graph this graph is now divided into two graphs the peritoneal region in the frontal region and you are trying to compare them based on event the stimulus event and the Q event the same conclusions are now seamlessly presented in the say in a very confined space in two columns so what I do I just try to show this row region so this is how you get instead of column if I do row this is how the graphs are presented and if you do column back so this is what you get right so this is it and maybe I can also test this but doesn't work the bar so you can use the replot function to present the graphic that you are trying to present a on a particular categorical variable and replicate it across the categorical variables and rows and columns so that you can make a quick comparison and it will be very easy for the audience to decipher trends so the same graph would require more mental effort on part of the audience to decipher as compared to this particular graph which is much more easier they can see okay in peritoneal region stimulus is much more higher than the Q as compared to the frontal region and again between peritoneal and frontal peritoneal stimulus on an average as lower as similar signal strength as compared to the peritoneal region and the Q is also aware as the Q is opposite Q is in both Q is also lower in the frontal region as compared to the peritoneal region but the differences are lesser as compared to the stimulus event so these kind of conclusions can be immediately drawn though though you can also draw them from this particular graphic but it would require more cognitive ability on part of your audience so always remember try to reduce the cognitive ability of your audience and by presenting more aesthetically pleasing graphs and you can use the re-plot feature in seaborn to achieve the same function okay thank you.

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